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Dual Laser-Aided UAV Motion Compensation for Vision-Based Displacement Measurement of High-Speed Railway Bridge

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) offer mobility and flexibility for vision-based structural health monitoring but are susceptible to motion-induced errors in displacement tracking. This study proposes a dual-laser compensation method in which two fixed laser beams, projected near the target to serve as reference points, are tracked to estimate UAV translation and rotation via affine transformation. By removing this motion component from the measured signal, the actual structural displacement can be obtained. Validation through both laboratory and field tests on a railway bridge demonstrates the effectiveness of the proposed method.

1. Introduction

Unmanned Aerial Vehicles (UAVs) have emerged as effective tools for structural health monitoring (SHM), offering rapid deployment, large-area coverage, and vision-based non-contact displacement monitoring [1]. However, UAV mobility introduces motion errors caused by wind, control inaccuracies, and navigation noise, which complicate the separation of true structural motion from camera movement [2].

Existing correction approaches include applying high-pass filtering to vibration data [3], using pixels assumed to be stationary within the frame [4], and employing sensors to measure camera vibrations [5]. However, these methods have inherent limitations; for example, stationary pixels may not exist in certain scenarios, such as in mid-span frames of a bridge [6], and additional onboard sensors increase system complexity and payload weight. Moreover, filtering techniques can inadvertently attenuate low-frequency structural responses that are critical for evaluating bridge behavior under operational loads.

In this study, we propose a novel dual-laser-aided vision correction method. Two fixed, ground-based laser points are projected onto the bridge, and their apparent motion in the UAV video is used to estimate

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an affine transformation representing UAV-induced movement. This approach enables simultaneous correction of both translational and rotational camera motions. By removing the motion component from the observed target displacement, the actual structural displacement can be obtained. The performance of the proposed method is validated through laboratory tests and field experiments on operational railway bridges, demonstrating its potential for accurate, non-contact displacement monitoring in challenging field conditions.

2. Theoretical background

2.1. Computer Vision-Based Displacement Measurement with Sub-Pixel Refinement

In this study, the template matching method [7] ×ÁÓ ÕÓÅÄ ÔÏ ÔÒÁÃËÉÎÅÁÓÇÂûÓ Í ÏÖÅ Í ÅÎÔÉÎ ÔÈÅ ÖÉÄÅÏ frames. Zero-mean normalized cross-correlation (ZNCC) was applied, as it is insensitive to offset and linear changes in illumination [8].

$$C_{\mathrm{ZNCC}} = \sum_{i=-M}^{M} \sum_{j=-N}^{N} \frac{\left(f\left(x_i, y_j\right) - f_m\right)\left(g\left(x'_i, y'_j\right) - g_m\right)}{\Delta f \, \Delta g} \quad (1)$$

Where $f\left(x_i, y_j\right)$ and $g\left(x'_i, y'_j\right)$ represent the image intensity values in the template region and the new frame, respectively; f_m and g_m are the average intensity values in the template region and the new frame; Δf and Δg denote the standard deviations of the intensity values in the template region and the new frame.

The integer-pixel location obtained from the ZNCC matrix provides only a coarse estimate of the target displacement. To further improve displacement measurement accuracy, sub-pixel tracking using the quadratic interpolation method [9] was implemented. The normalized cross-correlation matrix around the best match is locally approximated using a second-order (quadratic) surface:

$$S\left(x, y\right) = ax^2 + bxy + cy^2 + dx + ey + f \quad (2)$$

To fit this surface, a 3×3 neighborhood around the maximum point $\left(x_i, y_j\right)$ in the similarity matrix is selected, yielding 9 known similarity values. Then the subpixel maximum $\left(x_s, y_s\right)$ of the surface is found by setting the gradients to zero:

$$\frac{\partial S}{\partial x} = 2ax + by + d = 0 \quad (3)$$

$$\frac{\partial S}{\partial y} = 2cy + bx + e = 0 \quad (4)$$

Solving this linear system yields the refined displacement in sub-pixel units relative to the integer peak location.

$$x_s = \frac{2ce - bd}{b^2 - 4ac} \quad (5)$$

$$y_s = \frac{2ad - be}{b^2 - 4ac} \quad (6)$$

The total displacement is the integer-pixel estimate plus the sub-pixel correction in Equations (5)-(6). An illustration of the sub-pixel refinement using quadratic interpolation is shown in Figure 1. Figure 1a presents the coarse ZNCC values on a 3×3 pixel grid, and Figure 1b shows the quadratic surface fitted to this 3×3 neighborhood. The new extremum, marked by a triangle, indicates the refined sub-pixel best-matching location.

a) b)

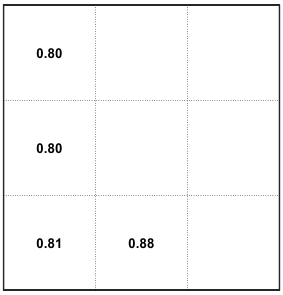

Figure 1. Quadratic peak refinement. (a) 3×3 correlation values, (b) fitted quadratic surface.

2.2. Dual-Laser Correction Using Affine Transformation

During measurement, the UAV experienced significant motions, including both translation and rotation, as shown in Figure 2. In earlier research [10], a single laser beam was projected from a fixed ground location onto the structure, and the apparent motion of the laser spot was subtracted from the measured target displacement to estimate the true structural displacement.

$$D_{\mathrm{true}} = D_{\mathrm{measured}} - D_{\mathrm{laser}} \quad (7)$$